

## The Scenario: Setting Up the Café Network

To handle high-speed IPL live streaming on a main screen while simultaneously maintaining low-latency, lag-free paths for multiplayer gaming matches, the network must be built hierarchically.

1. **Connecting to the ISP Line:** You would connect the incoming Internet Service Provider (ISP) line (typically coming out of the fiber optic modem) directly into the **Router's WAN (Internet) port** first.
2. **Connecting the Gaming PCs:** From one of the **Router's LAN ports**, you run a single high-speed Ethernet cable down to an **8-port (or larger) Gigabit Network Switch**. Finally, you plug all 8 gaming PCs directly into the ports of that network switch using individual Cat6 Ethernet cables.

## Justification of Choices

### Why the Router Comes First

- **Gateway to the Outside World:** Your ISP provides a single public IP address. The router is the only device capable of taking that single internet connection and sharing it. It uses **NAT (Network Address Translation)** to assign unique, private IP addresses to every device inside the café.
- **Traffic Prioritization (QoS):** Live streaming an IPL match in 4K can hog massive bandwidth. A router allows you to configure **Quality of Service (QoS)** protocols. You can set a rule to guarantee low-latency bandwidth to the multiplayer gaming PCs so their ping doesn't spike when the stream hits a high-action moment.

### Why a Switch is Used for the Gaming PCs

- **Eliminating Latency and Collisions:** While a home router might have 4 LAN ports on the back, it lacks the physical capacity for 8 PCs, and its internal processor isn't optimized to handle heavy, simultaneous local data traffic. A dedicated switch creates a dedicated, full-duplex pathway for each PC. This means Player 1 and Player 2 can exchange data at lightning speed without colliding or slowing down the rest of the network.
- **Hardware-Level Speed:** The switch reads physical **MAC addresses** right at the hardware level (Layer 2). When the local gaming server sends data to Player 5, the switch flips the data directly to Player 5's port instantly, leaving the router free to focus entirely on processing the external IPL stream.